

Baldyvis
Secondary College

Resource Section (Calculators and 1 x A4 (both sides) notes allowed)

/ 55 marks

5 marks

a. 100 ml
b. 1000 ml
c. 100 L
d. 1000 L

Volume

6. Convert the following:

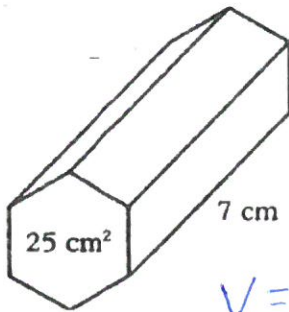
4 marks

- a. 2,400,000 cm³ to 2.4 m³ c. 8.3 cm³ to 8300 mm³
 b. 2.8 ML to 2800 000 L d. 0.61 KL to 610 L

7. Determine the volume of the following prisms and solid:

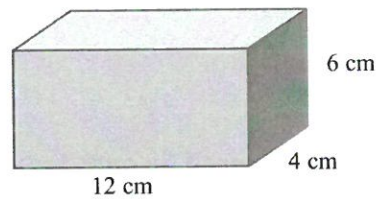
9 marks (2,2,2,3)

a.



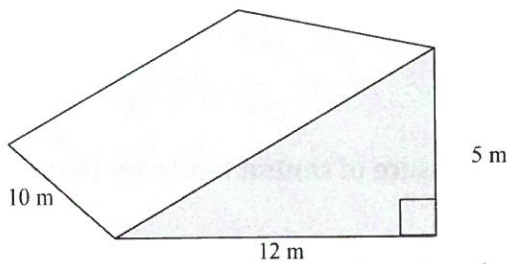
$$\begin{aligned} V &= \text{area of base} \times \text{length} \\ V &= 25 \times 7 \\ V &= 175 \text{ cm}^3 \end{aligned}$$

b.



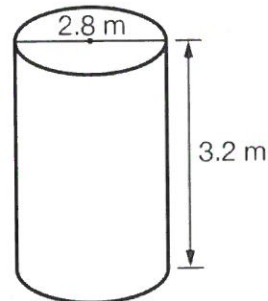
$$\begin{aligned} V &= l \times w \times h \\ &= 12 \times 6 \times 4 \\ &= 288 \text{ cm}^3 \end{aligned}$$

c.



$$\begin{aligned} V &= \frac{1}{2} \times \text{base} \times \text{height} \times \text{length} \\ &= \frac{1}{2} \times 12 \times 5 \times 10 \\ &= 300 \text{ m}^3 \end{aligned}$$

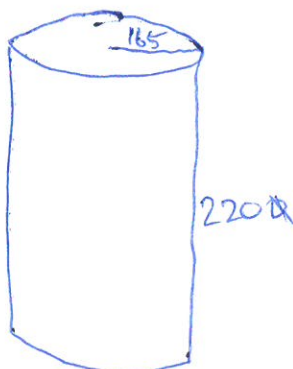
d.



$$\begin{aligned} V &= \pi r^2 h \\ &= \pi \times 1.4^2 \times 3.2 \\ &= 19.7 \text{ m}^3 \end{aligned}$$

8. A water tank has a radius of 165cm and a height of 2.2 m. How much water will the tank hold to the nearest litre?

4 marks



$$\begin{aligned} V &= \pi r^2 h \\ &= \pi \times 165^2 \times 2.2 \\ &= 18\,816\,569 \text{ cm}^3 \\ C &= 18\,816\,569 \text{ mL} \\ &= 18\,817 \text{ L} \end{aligned}$$

Scientific Notation

9. Write the following as numerals:

2 marks

a. $3 \times 10^4 =$ 30 000

b. $7.58 \times 10^{-5} =$ 0.0000758

10. Write the following in scientific notation:

2 marks

a. 8, 000, 000 = 8×10^6

b. 0.009802 = 9.802×10^{-3}

11. Solve the following. Write your answer in scientific notation.

4 marks

a. $(6.3 \times 10^2)(9.8 \times 10^3)$

$$\begin{aligned} & 6.3 \times 9.8 \times 10^2 \times 10^3 \\ &= 61.74 \times 10^5 \\ &= 6.174 \times 10^6 \end{aligned}$$

b. The amount of ants (8×10^{-6} km) that could fit around the equator (4×10^4 km)

$$\begin{aligned} & 4 \times 10^4 \div 8 \times 10^{-6} \\ &= 4 \div 8 \times 10^{4-(-6)} \\ &= 0.5 \times 10^{10} \\ &= 5 \times 10^9 \end{aligned}$$

Statistics

12. Frida collected surveyed her class and the class next door and recorded the percentage each person received on their latest maths test.

6 marks

a. Is this an example of primary or secondary data?

Primary

b. She presented the data in the graph below. What specifically is this type of graph called?

Back-to-Back
Stem & Leaf

Class A

1	5	4	2						
1	6	8	5	4					
5	7	7	9	6	1	6			
6	6	7	8	9	7	2	5	6	6
1	2	8	0	0	4	8	9		
1	9	3	5	6	7				

Class B

c. She thought the best way to analysis her data was with the median. What is the difference between the medians for the two classes? **Show your working.**

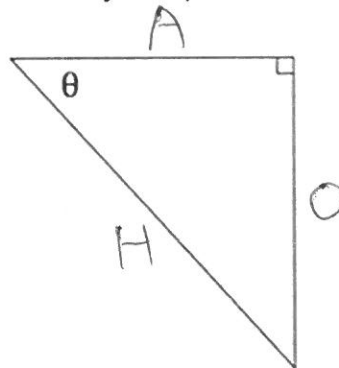
$$80 - 69 = 11$$

d. Which class do you think did better in the test and explain your answer by commenting on the shape of the graph for each class?

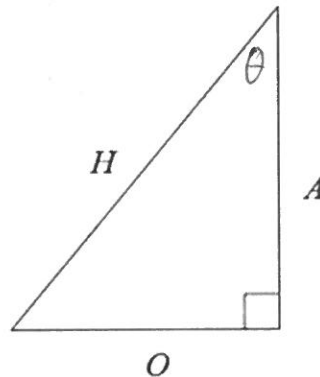
Class B because their results are skewed
Negatively (the left tail is longer)

Trigonometry

12. Label the sides (Hypotenuse, Opposite and Adjacent) of the following right angled triangle. **1 mark**

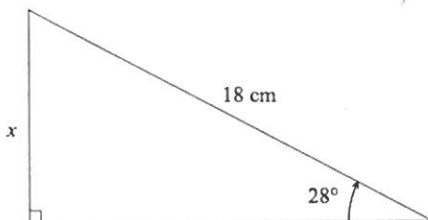


13. Label the angle (θ) in the correct place on the following right angled triangle. **1 mark**



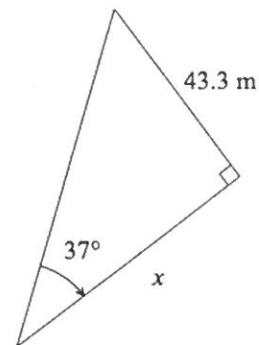
14. Calculate the length of x for each of the following right angled triangles to 2 decimal places. **4 marks**

a.



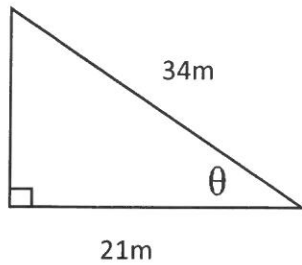
$$\begin{aligned}\sin \theta &= \frac{O}{H} \\ \sin 28 &= \frac{x}{18} \\ x &= 18 \times \sin 28 \\ x &= 8.45 \text{ cm}\end{aligned}$$

b.



$$\begin{aligned}\tan \theta &= \frac{O}{A} \\ \tan 37 &= \frac{43.3}{x} \\ x &= 43.3 \div \tan 37 \\ x &= 57.46 \text{ m}\end{aligned}$$

15. Find the size of the missing angle in the following right angled triangle to the nearest degree. 2 marks

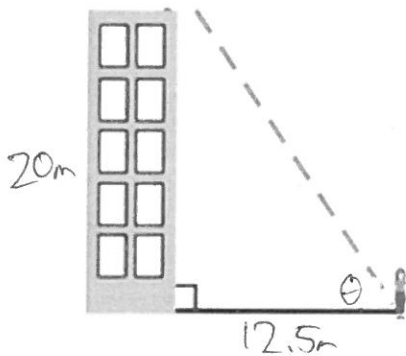


$$\cos \theta = \frac{A}{H}$$

$$\theta = \cos^{-1} (21 \div 34)$$

$$\theta = 52^\circ$$

16. Amir stands 12.5 m from the base of a building, looking up at the top of the building. If the building is 20m high, what angle is formed by the person's line of sight and the ground? 3 marks



$$\tan \theta = \frac{O}{A}$$

$$\tan \theta = \frac{20}{12.5}$$

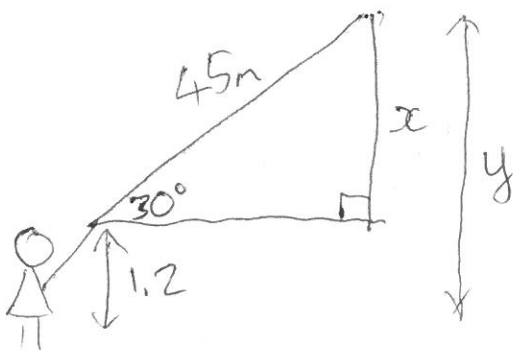
$$\theta = \tan^{-1} (20 \div 12.5)$$

$$\theta = 58^\circ$$

The angle formed is 58°

Extension Questions

17. A person flying a kite holds the string 1.2 m above level ground, at an angle of elevation of 30° . If the string is 45 m long, what is the height of the kite above ground level. (Sketch a diagram of the scenario) 4 marks



$$\sin \theta = \frac{O}{H}$$

$$\sin 30 = \frac{x}{45}$$

$$\sin x = 45 \times \sin 30$$

$$x = 22.5$$

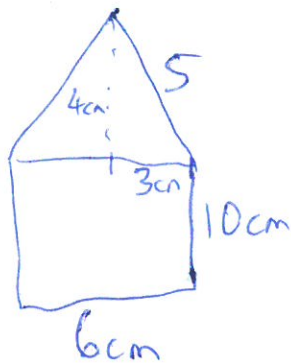
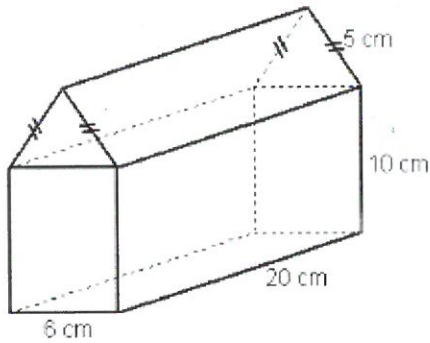
$$y = 22.5 + 1.2$$

$$y = 23.7m$$

The kite is 23.7m above the ground

18. Find the volume of the composite solid below

4 marks



$$b^2 = c^2 - a^2$$

$$b^2 = 5^2 - 3^2$$

$$b = \sqrt{25 - 9}$$

$$b = 4 \text{ cm}$$

$$\text{Area of front} = (b \times h_1) + (b \times h_2 \times \frac{1}{2})$$

$$= 6 \times 10 + 6 \times 4 \times \frac{1}{2}$$

$$= 60 + 12$$

$$= 72 \text{ cm}^2$$

$$\text{Volume} = A_{\text{Front}} \times l$$

$$= 72 \times 20$$

$$= \underline{1440 \text{ cm}^3}$$

End of Test